

The optimal estimator (the Kalman filter) is given as

$$\begin{aligned}
 P_e(0) &= E[(x(0) - \hat{x}(0))(x(0) - \hat{x}(0))^T] \\
 \dot{P}_e &= AP_e + P_eA^T - P_eC^TR^{-1}CP_e + Q \\
 K_e &= P_eC^TR^{-1} \\
 \dot{\hat{x}} &= A\hat{x} + K_e(y - C\hat{x})
 \end{aligned} \tag{8.153}$$

The differential Riccati equation for the optimal estimator is integrated forward in time from its initial condition $P_e(0)$.

The optimal control problem begins with the system

$$\dot{x} = Ax + Cu \tag{8.154}$$

where u is the control variable. The finite-time optimal control problem tries to find the control u that minimizes the cost function

$$J_c = x^T \phi x|_{t_f} + \int_0^{t_f} (x^T Q x + u^T R u) dt \tag{8.155}$$

ϕ , Q , and R (which are assumed to be symmetric positive definite matrices) provide user-specified weighting in the performance index. The optimal controller is given as

$$\begin{aligned}
 P_c(t_f) &= \phi(t_f) \\
 \dot{P}_c &= -A^T P_c - P_c A + P_c C R^{-1} C^T P_c - Q \\
 K_c &= R^{-1} C^T P_c \\
 u &= -K_c x
 \end{aligned} \tag{8.156}$$

The differential Riccati equation for the optimal control problem is integrated backward in time from the final condition $P(t_f)$. Note the relationships between the optimal estimation solution of Equation (8.153) and the optimal control solution of Equation (8.156). The differential Riccati equations have the same form, except they are negatives of each other, and A and C are replaced by their transposes. The estimator gain K_e and the controller gain K_c have very similar forms. The Q and R covariance matrices in the estimation problems have duals in the cost function weighting matrices of the optimal control problem.

The dual relationship between the estimation and control problems was noted in the very first papers on the Kalman filter [Kal60, Kal61]. Since then, it has been used many times to extrapolate results known from one problem to obtain new results for the dual problem.

8.6 SUMMARY

In this chapter, we derived the continuous-time Kalman filter by applying a limiting argument to the discrete-time Kalman filter. However, just as there are several ways to derive the discrete-time Kalman filter, there are also several ways to derive the continuous-time Kalman filter. Kalman and Bucy's original derivation [Kal61] involved the solution of the Wiener-Hopf integral equation. Another derivation is provided in [Joh69].

We have seen that the differential and algebraic Riccati equations are key to the solution of the continuous-time Kalman filter. The scalar version of what is now known as the Riccati equation was initially studied by such mathematical luminaries as James Bernoulli and John Bernoulli in the 1600s, and Jacopo Riccati, Daniel Bernoulli, Leonard Euler, Jean-le-Rond d'Alembert, and Adrien Legendre in the 1700s. The equation was first called "Riccati's equation" by d'Alembert in 1763 [Wat22]. Jacopo Riccati originally entered the University of Padua in 1693 to study law, but he found his true calling when his astronomy professor, Stefano Angeli, inspired him to study math. Additional technical discussion of Riccati equations can be found in many places, including [Rei72, Lan95, Abo03]. An account of Riccati equations with indefinite quadratic terms is given in [Ion99]. Interesting historical background to the Riccati equation can be found in [Wat22, Bit91].

The continuous-time Kalman filter applies to systems with continuous-time white noise in the both the process and measurement equations. Continuous-time white noise is nonintuitive because it has an infinite correlation with itself at the present time, but zero correlation with itself when separated by arbitrarily small nonzero times. However, continuous-time white noise is a limiting case of discrete-time white noise, which is intuitively acceptable. Therefore, continuous-time white noise can be accepted as an approximation to reality. This corresponds to many other approximations to reality that we accept at face value (e.g., our mathematical system model is an approximation to reality, and our infinite-precision arithmetic is an approximation to reality).

The continuous-time Kalman filter applies regardless of the statistical nature of the noise, as long it is zero-mean. That is, the Kalman filter is optimal even when the noise is not Gaussian. The Kalman filter was extended in this chapter to systems with correlated process and measurement noise, and with colored measurement noise. The steady-state Kalman filter provides near-optimal estimation performance at a small fraction of the computational effort of the time-varying Kalman filter. The steady-state Kalman filter is identical to the Wiener filter of Section 3.4, and has an interesting dual relationship to linear quadratic optimal control.

PROBLEMS

Written exercises

8.1 Suppose you have two discrete-time systems with identity transition matrices driven with stationary zero-mean white noise. The first system has a sample period of T , and the second system has a sample period of T/n for some integer $n > 1$. The noise in the first system has a covariance of Q . What should the covariance of the noise in the second system be in order for both states to have the same covariance at times kT ($k = 0, 1, 2, \dots$)?

8.2 Show that for a general time-varying matrix $Y(t)$, if $\dot{Y} = AY + YA^T$, where A is a constant matrix, then $Y(t) = \exp(At)Y(0)\exp(A^T t)$.

8.3 Suppose you have a third-order Newtonian system with

$$\begin{aligned} A &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \\ C &= \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \\ Q &= \begin{bmatrix} 2 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \\ R &= 1 \end{aligned}$$

with $P(0) = I$.

- What is the rank of $\dot{P}(0)$? How much computational savings in integration effort can be obtained by using the Chandrasekhar algorithm to find the Kalman gain for this system?
- Find M_1 and M_2 such that $\dot{P}(0) = M_1 M_1^T - M_2 M_2^T$.

8.4 Show that if S is upper triangular, then $\dot{S}$ and S^{-1} are also upper triangular.

8.5 Show that the product of upper triangular matrices is another upper triangular matrix.

8.6 Find the steady-state solution of the differential Riccati equation for a scalar system. Show from your solution how the steady-state solution changes with A , C , Q , and R , and give intuitive explanations.

8.7 Consider the system of Example 8.3 except with process noise that has a covariance of $\text{diag}(0, q)$. Find an analytical expression for the steady-state estimation-error covariance.

8.8 Show that if $a_1 \neq a_2$ and $q_{12} \neq 0$ in the system of Example 8.5, then (A, C) is detectable and (A, G) is stabilizable for all matrices G such that $GG^T = Q$.

8.9 Show that if $a_1 = a_2 < 0$, $q_{12} \neq 0$, and $|Q| = 0$ in the system of Example 8.5, then (A, C) is detectable and (A, G) is stabilizable for all matrices G such that $GG^T = Q$.

8.10 Show that if $a_1 = a_2 > 0$, $q_{12} \neq 0$, and $|Q| = 0$ in the system of Example 8.5, then (A, C) is detectable and (A, G) is controllable on the imaginary axis, but (A, G) is not stabilizable for all matrices G such that $GG^T = Q$.

Computer exercises

8.11 Consider the discrete-time system $x_{k+1} = x_k + w_k$ with the initial condition $x_0 = 0$. The sample time is T and the variance of the zero-mean process noise w_k is equal to $2T$. Simulate the system a few thousand times for 10 s with: (a) $T = 0.5$ s; (b) $T = 0.4$ s; (c) $T = 0.2$ s. Use the value of x_k at $t = 10$ s to obtain a statistical estimate of $P(10) = E[x^2(10)]$.

- What is your estimate of $P(10)$ for the three sample times given?
- What is the analytically derived value for $P(10)$?

8.12 Consider the continuous-time scalar system

$$\begin{aligned}\dot{x} &= -x + w \\ y &= x + v\end{aligned}$$

where $w(t)$ and $v(t)$ are continuous-time white noise with variances $Q_c = 2$ and $R_c = 1$ respectively. Design a continuous-time Kalman filter to estimate x .

- a) What is the theoretical steady-state variance of the estimation error?
- b) Simulate the system for 1000 s with discretization step sizes of 0.4, 0.2, and 0.1 s. What are the resulting experimental estimation-error variances?

8.13 Simulate the system of Problem 8.7 for 10 seconds with $q = 2$ and $R = 3$. Plot the elements of the estimation-error covariance matrix as a function of time. Compare the experimental RMS estimation errors when using a time-varying Kalman gain and a constant Kalman gain.

8.14 Repeat Problem 8.13 using the correlated noise filter when the process noise that affects the second state is equal to the measurement noise. How much do the estimation-error variances decrease due to the correlation between the two noise terms?

8.15 Consider the system of Example 8.5 with $R = I$.

- a) Integrate the Riccati equation with $a_1 = 1$, $a_2 = 2$, $q_{11} = q_{12} = q_{22} = 1$, and $P(0) = I$. Plot the Riccati equation solution as a function of time and verify that its steady-state value matches the results of Equation (8.139) and MATLAB's CARE function.
- b) Integrate the Riccati equation with $a_1 = a_2 = -1$, $q_{11} = 1$, $q_{12} = 2$, $q_{22} = 4$, and $P(0) = I$. Plot the Riccati equation solution as a function of time and verify that its steady-state value matches the results of Equation (8.140) and MATLAB's CARE function.
- c) Integrate the Riccati equation with $a_1 = a_2 = 1$, $q_{11} = 1$, $q_{12} = 2$, $q_{22} = 4$, and $P(0) = I$. Plot the Riccati equation solution as a function of time and verify that its steady-state value matches the results of Equation (8.139) and MATLAB's CARE function.
- d) Integrate the Riccati equation with $a_1 = a_2 = 1$, $q_{11} = 1$, $q_{12} = 2$, $q_{22} = 4$, and $P(0) = 0$. [Note that this is the same as part (c) except for $P(0)$.] Plot the Riccati equation solution as a function of time and verify that its steady-state value matches the results of Equation (8.140). Does it match the results of MATLAB's CARE function? Does it result in a stable steady-state Kalman filter?